SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

SRM Institute of Science and Technology · Academic Year 2026

INSTITUTIONAL MASTER EDITION

Official Production Portal: Hosted on Vercel Cloud `https://electifysrm.vercel.app`

Institutional Operations & Architecture Manual

Institutional Master Manual: Comprehensive architectural, functional, and operational reference for SRMIST academic evaluators, department heads, and system administrators.

1. Production Cloud Endpoints

Layer	Technology	Live Production Endpoint
Frontend Portal	Next.js 16 (React 19, Tailwind v4)	`https://electifysrm.vercel.app`
Student Sign-Up / Join	Next.js Server Actions	`https://electifysrm.vercel.app/join`
Faculty Sign-In	Next.js & JWT Auth	`https://electifysrm.vercel.app/login`
Backend REST API	NestJS 11 (Express, Swagger)	`https://electifysrm.vercel.app/api` (Docs: `/api/docs`)
Database	Supabase PostgreSQL 16	Transaction Pooler (`aws-1-eu-central-1`)

2. University Role Credentials & Authority

Role	Account Identifier	Password / Access Method	Authority Scope
System Admin	`srmelectify@gmail.com`	`Electify@2026`	Root control, maintenance mode, system audit trail.
Course Coordinator	`sabenabanu@srmist.edu.in`	`Electify@2026`	Elective baskets, seat limits, window timers, surveillance, master reports.
Class Tutor	`angelb1@srmist.edu.in`	`Electify@2026`	Class code generation, section rosters, student PIN reset authorization.
Student	SRM Register Number	Student Configured via `/join`	Subject explorer preview, atomic seat claim, PDF receipt.

3. Technical Concurrency & Safeguards

- **PostgreSQL Atomic Row-Locking:** Seat reservations use `SELECT ... FOR UPDATE` within ACID transactions. Zero waitlists and zero overbooking.
- **Role-Based Session Sentinel:** Strict single-device enforcement for students to eliminate multi-tab sniping; multi-device access enabled up to **3 active devices** simultaneously for Class Tutors, Course Coordinators, and System Administrators.

- **NTP Clock Synchronization:** Client countdown timers are synchronized to server NTP time to eliminate client-side device tampering.
- **Tamper-Evident Receipts:** Formatted as `ELC-2026-EVEN-XXXXX` with cryptographic verification signatures.

4. Automated SQA Verification (Zero-Defect Status)

> **Automated Quality Assurance: 25 Tests Passed / 0 Failed (100% Pass Rate)**
> Execute anytime with `npm run test:sqa` to verify Supabase connectivity, 27 schema tables, multi-device RBAC policies, Class Tutor & Coordinator actions, and REST endpoints.